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Lecture proposal: LLM classifiers / "Classy LLMs"
September 23, 2026

Learning objectives

We hope students will come away from our seminar with three understandings:

1. LLMs can be used for classification problems, both binary and multiclass, and are currently used in content moderation
2. LLM classifiers present strong benefits over more traditional statistical methods when handling text data, and through human-readable prompts allows for a unique window into how the model evaluates its inputs
3. LLM classifier prompts can be refined iteratively to in essence "train" the policy text, both with human-in-the-loop and self-contained approaches

LLM classifiers are a relatively new and understudied technology, and are incredibly useful in allowing for human-configurable classifiers that are more adaptable and less of a black box than many traditional statistical approaches. We expect that, as LLM classifiers become more popular both in industry and academia, our learning this content as prep and exposing the class to it will be useful for our development as data scientists.

Core readings

Likely Wang, Pang, and Lin (2024) and Chakrabarti et al. (2025). The first builds background on using LLM classifiers for text data, and the second applies LLM classifiers to content moderation problems.

Class outline

We expect to cover three topics; each section will begin with a short lecture and transition into more workshop-style small group discussions.

1. Introduction to classifiers and LLMs (∼45min)
    - Presenters: Alex
    - Source material: Google for Developers (2026), GeeksforGeeks (2026), Kumar (2024)

- Lecture idea: Refresh on classifiers (map from feature set to [0,1], example using logistic regression, thresholds, precision, recall, F1), introductory LLM architecture (transformers).
- Activity/workshop idea: Create an interactive demo following the steps of a transformer.

2. LLMs as classifiers (∼1hr30min)
   - Presenters: Luc and Nolan
   - Source material: You et al. (2025), Dai et al. (2025), Lam (2026), Wang, Pang, and Lin (2024), Henrichsen and Krebs (2025), Pangakis and Wolken (2024)
   - Lecture idea and activity:
     - Nolan: Pros and Cons of different models: open source vs paid models; Possible demo: showing the differences between an open source and an industry paid model on the same classification project
     - Luc: How to use LLMs for semi-supervised learning (label extraction + classification)
3. LLM classifiers in action: policy refinement and application to content moderation (∼45min)
   - Presenters: Rae
   - Source material: Kolla et al. (2024), Chakrabarti et al. (2025)
   - Lecture idea: Theory of refining LLM classifier prompts ("policies") with iterative loops to maximize F1. Overview of applications to content moderation on Reddit and Wikipedia.
   - Activity/workshop idea: Demo of classifier iteration and small group discussions about potential use cases of LLM classifiers.

Dates

We can present on all dates except 10/28.

References

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Kumar, Pranjal (Aug. 18, 2024). "Large Language Models (LLMs): Survey, Technical Frameworks, and Future Challenges". In: *Artificial Intelligence Review* 57.10, p. 260. ISSN: 1573-7462. DOI: 10.1007/s10462-024-10888-y. URL: https://doi.org/10.1007/s10462-024-10888-y (visited on 09/21/2026).

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Pangakis, Nicholas and Samuel Wolken (June 25, 2024). *Knowledge Distillation in Automated Annotation: Supervised Text Classification with LLM-Generated Training Labels*. Version 1. DOI: 10.48550/arXiv.2406.17633. arXiv: 2406.17633 [cs.CL]. URL: http://arxiv.org/abs/2406.17633 (visited on 09/21/2026). Pre-published.

Wang, Zhiqiang, Yiran Pang, and Yanbin Lin (May 17, 2024). *Smart Expert System: Large Language Models as Text Classifiers*. Version 1. DOI: 10.48550/arXiv.2405.10523. arXiv:

2405.10523 [cs.CL]. URL: http://arxiv.org/abs/2405.10523 (visited on 09/21/2026). Pre-published.

You, Doohee et al. (Aug. 22, 2025). *LLM-as-classifier: Semi-Supervised, Iterative Framework for Hierarchical Text Classification Using Large Language Models*. DOI: 10.48550/arXiv.2508.16478. arXiv: 2508.16478 [cs.CL]. URL: http://arxiv.org/abs/2508.16478 (visited on 09/21/2026). Pre-published.